

1. Regression vs Classification (First Step in ML Model Building)

Before building a machine learning model, the **first decision** is identifying the **type of problem**.

Two major supervised learning problem types:

```
Supervised Machine Learning
|
|---- Regression
|---- Classification
```

2. Regression

Definition

Regression predicts **continuous numerical values** such as price, temperature, or sales.

In simple words:

Regression = Predict a number.

Examples

Problem	Output
House price prediction	75,00,000
Temperature forecast	32.5°C
Stock price	1500
Sales prediction	1200 units

Example Dataset

Size	Bedrooms	Location Score	Price
1200	2	8	70L
1500	3	9	95L
2000	4	10	1.4Cr

Goal:

Predict Price

for a **new house**.

Example Output

Input:

Size = 1700
Bedrooms = 3
Location Score = 9

Prediction:

Price = 1.05 Cr

3. Classification

Definition

Classification predicts **categories or labels** such as spam/not spam or fraud/not fraud.

In simple words:

Classification = Predict a category.

Examples

Problem	Output
Spam detection	Spam / Not Spam
Disease diagnosis	Positive / Negative
Customer churn	Yes / No
Image recognition	Cat / Dog

Example Dataset

Age	Income	Credit Score	Loan Default
25	40K	650	No

45	80K	720	No
30	25K	580	Yes

Goal:

Predict Loan Default (Yes / No)

4. Key Differences

Feature	Regression	Classification
Output	Continuous value	Category
Example	House price	Spam email
Target variable	Numeric	Categorical
Prediction	Number	Label

5. How to Identify the Problem Type

Ask:

What is the output?

Output type	Problem type
Number	Regression
Category	Classification

Examples:

Problem	Type
Predict house price	Regression
Predict customer churn	Classification
Predict temperature	Regression
Detect fraud	Classification

6. Machine Learning Model Building Steps

Typical ML pipeline:

- 1 Problem Definition
- 2 Data Collection

3 Data Understanding / EDA
4 Data Cleaning
5 Feature Engineering
6 Train-Test Split
7 Model Training
8 Model Evaluation
9 Model Deployment

These steps are the **same for regression and classification**.

7. Step 1 — Problem Definition

Define:

- Business objective
 - Input features
 - Target variable
 - Problem type
-

Example 1 (Regression)

Problem:

Predict house price

Features:

- house size
- bedrooms
- location
- age of property

Target:

price

Example 2 (Classification)

Problem:

Predict customer churn

Features:

- monthly charges
- contract type
- tenure

Target:

churn (Yes/No)

8. Step 2 — Data Collection

Data sources:

- databases
- APIs
- logs
- sensors
- CSV files

Example dataset:

house_data.csv

Python Example

Pseudocode

```
Load dataset
Display sample rows
Check dataset structure
```

Code

```
import pandas as pd

df = pd.read_csv("house_data.csv")

df.head()
```

Output

Size	Bedrooms	Location	Price
1200	2	Pune	70L

9. Step 3 — Exploratory Data Analysis (EDA)

Goal:

Understand data patterns.

Things to check

- data distribution
 - correlations
 - missing values
 - outliers
-

Example

Check summary statistics.

Pseudocode

```
Check dataset shape
Check summary statistics
Check missing values
```

Code

```
df.shape
df.describe()
df.isnull().sum()
```

Example Insight

Average house price = 95L
Max bedrooms = 5

10. Step 4 — Data Cleaning

Real-world data contains problems:

- missing values
 - duplicates
 - incorrect formats
-

Example Problem

Dataset:

Age	Salary
25	50000
30	NULL
28	60000

Fix Missing Values

Code

```
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
```

11. Step 5 — Feature Engineering

Feature engineering improves model performance.

Examples:

Raw Data	Engineered Feature
Date of birth	Age
Transaction history	Avg spending
Address	Region

Example

Convert categorical data.

Pseudocode

Convert categorical variable into numeric

Code

```
df = pd.get_dummies(df, columns=['Location'])
```

Example result:

Size	Bedrooms	Location_Pune	Location_Mumbai
1200	2	1	0

12. Step 6 — Train-Test Split

Split dataset into:

Training Data → learn patterns

Testing Data → evaluate model

Typical split:

80% training

20% testing

Pseudocode

Split dataset into X (features) and y (target)

Split into train and test

Code

```
from sklearn.model_selection import train_test_split
```

```
X = df.drop("Price", axis=1)
```



```
y = df["Price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

13. Step 7 — Model Training

Now the model learns patterns from training data.

Example:

Model learns relationship between features and target

Example pattern learned:

Larger house → higher price
Better location → higher price

Pseudocode

```
Initialize model
Train model on training data
```

Code

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)
```

14. Step 8 — Prediction

Once trained, the model can predict new values.

Pseudocode

```
Predict values for test data
```

Code

```
predictions = model.predict(X_test)

print(predictions[:5])
```

Example Output

```
[8700000, 9200000, 10100000]
```

Predicted house prices.

15. Step 9 — Model Evaluation

Evaluate model performance.

Different metrics for **regression vs classification**.

Regression Metrics

Examples:

Metric	Meaning
MAE	Average error
MSE	Squared error
RMSE	Root error

Code

```
from sklearn.metrics import mean_absolute_error

mean_absolute_error(y_test, predictions)
```

Example output:

```
MAE = 120000
```

Meaning:

Prediction error = ₹1.2 lakh.

Classification Metrics

Examples:

Metric	Meaning
Accuracy	Correct predictions
Precision	Correct positive predictions
Recall	Detection rate
F1 score	Balanced metric

Example Code

```
from sklearn.metrics import accuracy_score  
  
accuracy_score(y_test, predictions)
```

Example output:

Accuracy = 0.89

Meaning:

89% predictions correct

16. Step 10 — Model Deployment

After evaluation, deploy the model.

Deployment methods:

- API
 - Web application
 - Dashboard
 - Batch prediction
-

Example Architecture

User input → API → ML Model → Prediction

Example:

User enters house details
Model returns predicted price

17. Real End-to-End Example

Problem

Predict **customer churn**.

Dataset

Tenure	Monthly Charges	Support Calls	Churn
12	500	2	No
3	800	6	Yes

ML Pipeline

```
1 Define problem
2 Load data
3 EDA
4 Data cleaning
5 Feature engineering
6 Train-test split
7 Train model
8 Evaluate model
9 Deploy model
```

Example Prediction

Customer:

```
tenure = 2
monthly charges = 900
support calls = 7
```

Model output:

Churn = Yes
Probability = 0.85

18. Simple ML Model Architecture

Typical pipeline:

```
Data Source
      ↓
Data Cleaning
      ↓
Feature Engineering
      ↓
Train/Test Split
      ↓
Model Training
      ↓
Model Evaluation
      ↓
Deployment
```

19. Key Interview Tip

Always explain ML modeling in this order:

- 1 Problem type (Regression vs Classification)
 - 2 Data understanding
 - 3 Feature engineering
 - 4 Model training
 - 5 Model evaluation
 - 6 Deployment
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